1. Review title

Estimation of Mortality Hazard Ratios by COPD Stage Using Network Meta-Analysis: Deriving Stage-, Sex-, and Age-Specific Mortality for a German COPD Decision Model from a Systematic Review.

2. Original language title

NA

3. Anticipated or actual start date

12/06/2025

4. Anticipated completion date

30/04/2026

5. Stage of review at time of this submission

- Preliminary searches: **Completed**
- Piloting of study selection: **Completed**
- Formal screening: **Completed**
- Data extraction: **Completed**
- Risk of bias assessment: **Planned**
- Data analysis: **Ongoing**

6. Named contact

Tobias Niedermaier

7. Email

Tobias.niedermaier@ibe.med.uni-muenchen.de

8. Organisation

Institute for Medical Information Processing, Biometry, and Epidemiology (IBE), LMU Munich

9. Review team members and affiliations

- **Tobias Niedermaier, IBE, LMU Munich (Single reviewer with supervision by Rolf Holle, IBE, LMU Munich**

10. Funding sources/sponsors

No specific funding.

11. Conflicts of interest

None declared.

Review methods

12. Review question

Primary question:
Among adults classified by **GOLD 2007 spirometric COPD stages I–IV**, what are the **adjusted hazard ratios (HRs) for all-cause mortality** for each stage, preferably compared with **normal spirometry**, and how can these HRs be used to derive **age-, sex-, and stage-specific all-cause mortality rates for Germany**?

Secondary questions (if data permit):
Do pooled HRs differ by **study setting** (clinical cohorts vs population-based cohorts) and by **geographic region**?

13. Condition or domain being studied

Chronic obstructive pulmonary disease (COPD) severity classified by **GOLD 2007 spirometric stages (I–IV)** and **all-cause mortality**.

14. Type of review

Systematic review with meta-analysis (network meta-analysis of adjusted hazard ratios).
(Note for PROSPERO form limitation: This is a prognostic factor/exposure review; the intervention-review form is used because other review types are currently unavailable in PROSPERO.)

15. Participants/population

Adults (≥18 years) from population-based or clinical cohorts with spirometry-based classification into **normal spirometry** and/or **COPD GOLD 2007 stages I–IV**.

16. Exposure(s) / “Intervention(s)” (worded to fit the PROSPERO form)

Exposure/prognostic factor: COPD severity category defined by **GOLD 2007 stage I–IV** based on spirometry (FEV1/FVC and FEV1 % predicted).
No therapeutic intervention is evaluated.

17. Comparator(s)/control

Primary target comparator for harmonised synthesis: **normal spirometry**.
Other comparators accepted for inclusion: another single GOLD stage (e.g., GOLD I or GOLD II), which will be handled through network meta-analysis to obtain estimates on a common scale.

18. Main outcome(s)

All-cause mortality, extracted as **adjusted hazard ratios (HRs)** with **95% confidence intervals** by GOLD stage.

19. Additional outcome(s)

- Hazard ratios for mortality stratified by geographic region and by above-/below-median total case number.
- **Germany-specific annual all-cause mortality rates** stratified by **age group, sex, and GOLD stage**, derived by combining pooled HRs with official German mortality and population data.

20. Eligibility criteria (inclusion/exclusion)

Inclusion criteria

1. **Study type:** longitudinal observational studies (prospective or retrospective cohorts; registry-based cohorts) reporting time-to-event analyses.
2. **Publication type:** full-text original articles.
3. **Follow-up:** minimum follow-up duration **≥ 1 year**.
4. **Exposure definition:** COPD staged according to **GOLD 2007** (I–IV) using spirometry (typically FEV1/FVC <0.70 and FEV1 % predicted thresholds).
5. **Effect measure: adjusted HRs** (or estimates convertible to HRs) for **all-cause mortality**, with **95% CIs**, reported for at least one GOLD stage (I–IV).
6. **Reference group:** normal spirometry or another single GOLD stage (I–IV).
7. **Minimum adjustment for primary synthesis:**
    - **At least age and sex** if the cohort includes both sexes; and
    - **Smoking adjustment** (status or pack-years) is **required when normal spirometry is the reference**, but is **not required** when the reference group is another COPD stage (because smoking differences across stages are typically smaller than differences vs normal spirometry).
    - **Single-sex cohorts** are eligible even without sex adjustment.

Exclusion criteria

- Hospitalised-only or other clearly **high-risk clinical subgroups at recruitment** (e.g., recently discharged, dialysis) where mortality is dominated by acute/advanced comorbidity selection.
- Studies using **GOLD stage 0** (or similar mixed categories such as PRISm mixed into the reference) as reference.
- Conference abstracts, dissertations without full article, unpublished manuscripts.

- Non-longitudinal designs without time-to-event effect measures.

Definition of "normal spirometry"

Preferably FEV1/FVC ≥0.70 **and** FEV1 ≥80% predicted. If studies use slightly relaxed definitions (e.g., FEV1/FVC ≥0.70 only, or "no obstruction/no symptoms"), these will be included but flagged and potentially explored in sensitivity analyses.

21. Search strategy (databases, dates, other sources)

We will search **PubMed/MEDLINE** from **2004 to the search date**. Search terms will combine COPD, mortality/survival, longitudinal/cohort designs, staging (GOLD, spirometry), and effect measures (hazard ratio). We will also:

- Screen reference lists of included studies,
- Use forward citation searching of key included studies (where feasible).

The search will be **re-run shortly before final analyses** to capture recent publications.

22. Example search terms (brief)

("COPD" OR "chronic obstructive pulmonary disease") AND (mortality OR death OR survival) AND (cohort OR longitudinal OR prospective OR follow-up) AND (GOLD OR stage OR spirometry OR FEV1/FVC) AND ("hazard ratio" OR HR).

23. Study selection process

Two reviewers will independently screen titles/abstracts and then full texts. Disagreements will be resolved by consensus (or a third reviewer if needed). A PRISMA flow diagram will be produced.

24. Data extraction (selection and coding)

Data will be extracted into a standardised sheet (e.g., Excel). Extracted items will include:

- Study: author, year, country, cohort type (clinical vs population-based)
- Sample size, number of deaths (if available), follow-up duration
- Spirometry approach (post- vs pre-bronchodilator if reported)
- GOLD stage definitions used
- Definition of the comparator group, e.g. normal spirometry (if applicable)
- HRs and 95% CIs for each stage and reference group
- Covariates included in adjustment (age, sex, smoking, comorbidities, etc.)

If multiple models are reported, we will preferentially extract the model that balances our minimum adjustment rules with potential overadjustment (e.g., avoiding estimates adjusted for symptoms that are on the causal path between COPD and mortality).

25. Risk of bias (quality) assessment

Risk of bias will be assessed using the **Newcastle–Ottawa Scale (NOS)** for cohort studies (selection, comparability, outcome). We will report NOS scores (per item and summary scores) for each included study.

26. Strategy for data synthesis

Overview

Effect estimates will be synthesized as **log(HR)** with standard errors derived from reported 95% CIs. Because studies may use different reference categories (normal spirometry, GOLD I, GOLD II), we will use a **random-effects network meta-analysis (NMA)** to express all stage-specific effects on a **common comparator scale**, prioritizing **normal spirometry** as the anchor comparator.

Statistical model

- **Random-effects** NMA (frequentist framework; e.g., R package **netmeta**).
- Primary outputs: pooled HRs (and 95% CIs) for **GOLD I, II, III, IV vs normal spirometry**.

Handling of multiple estimates

If a single study reports multiple relevant HRs (e.g., sex-stratified) that originate from the same study population, we will pool them using fixed-effects meta-analysis.

Sensitivity analyses (planned)

- Restrict to studies using **post-bronchodilator** spirometry (when reported).
- Include studies without sex adjustment only in sensitivity (mixed-sex cohorts), not in primary synthesis.
- Alternative pooling restricted to studies that already use **normal spirometry** as reference (standard random-effects meta-analysis).

Small-study effects / reporting bias

Small-study effects will be explored using funnel plots and Egger-type approaches (interpretation cautious given heterogeneity and network structure).

27. Planned subgroup analyses

If sufficient data exist:

- **Clinical vs population-based cohorts**
- **Geographic region** (e.g., continent)
- **Study size** (e.g., above vs below median sample size)

28. Germany-specific derivation of mortality rates (key application)

Primary aim of the project: derive **annual all-cause mortality rates for Germany** by **age group, sex, and GOLD stage**.

Data sources

We will use official German **life table / mortality and population data** (German Federal Statistical Office, Destatis). To avoid distortion by the COVID-19 pandemic, we will preferentially use a **pre-pandemic year** (e.g., 2019) for the baseline.

Method

1. Obtain baseline age- and sex-specific probabilities of death (e.g., abridged life table 5-year age groups).
2. Convert baseline probabilities to **annual hazards**.
3. Multiply baseline hazards by pooled HRs for each GOLD stage (vs normal spirometry) to obtain **stage-specific hazards**.
4. Convert stage-specific hazards back to **annual mortality probabilities/rates**.
5. Uncertainty: derive 95% intervals for stage-specific mortality rates by propagating the **HR uncertainty** (baseline life table uncertainty is assumed negligible for national data).

Age range and open-ended interval

Because COPD is epidemiologically negligible at young ages, reporting will focus on ages **≥45 years**. The open-ended age group (e.g., **85+**) will be handled by aggregating component age bands where available, using weighted averaging based on approximate person-years.

29. Dissemination plans

Results will be submitted for publication in a peer-reviewed journal and will be presented in thesis/policy-model documentation. Extraction sheets and analysis code will be shared publicly where possible or made available on reasonable request.

30. Ethics and dissemination

No primary data will be collected; ethics approval is not required for secondary analysis of published studies and publicly available national statistics.

31. Other information (important for the PROSPERO “type of review” limitation)

This is a prognostic factor/exposure review (COPD severity stage as exposure) aiming to derive German age-, sex-, and stage-specific mortality rates. The PROSPERO “Intervention review” form is used because the dedicated prognostic review option is currently unavailable (“coming soon”).